

A Practical Guide to Explainable artificial intelligence

TOPIC -- EXPLAINABLE ARTIFICIAL INTELLIGENCE

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By just tracing through the dependency structure the problem solver can construct the following explanation: "Socrates died because he was mortal and drank poison, and all mortals die when they drink poison. Socrates was mortal because he was a man and all men are mortal. Socrates drank poison because he held dissident beliefs, the government was conservative, and those holding conservative dissident beliefs under conservative governments must drink poison." By the 1990s researchers began studying whether it is possible to meaningfully extract the non-hand-coded rules being generated by opaque trained neural networks. Researchers in clinical expert systems who created neural network-powered decision support for clinicians sought to develop dynamic explanations that allow these technologies to be more trusted and trustworthy in practice. In the 2010s public concerns about racial and other bias in the use of AI for criminal sentencing decisions and findings of creditworthiness may have led to increased demand for transparent artificial intelligence. As a result, many academics and organizations are developing tools to help detect bias in their systems. Marvin Minsky et al. raised the issue that AI can function as a form of surveillance, with the biases inherent in surveillance, suggesting HI (Humanistic Intelligence) as a way to create a more fair and balanced "human-in-the-loop" AI. Explainable AI has been recently a new topic researched amongst the context of modern deep learning. Modern complex AI techniques, such as deep learning, are naturally opaque. To address this issue, methods have been developed to make new models more explainable and interpretable. This includes layerwise relevance propagation (LRP), a technique for determining which features in a particular input vector contribute most strongly to a neural network's output, although this technique has been shown to suffer from several important issues. Other techniques explain some particular prediction made by a (nonlinear) black-box model, a goal referred to as "local interpretability". We still today cannot explain the output of today's DNNs without the new explanatory mechanisms, we also can't by the neural network, or external explanatory components. There is also research on whether the concepts of local interpretability can be applied to a remote context, where a model is operated by a third-party. There has been work on making glass-box models which are more transparent to inspection. This includes decision trees, Bayesian networks, sparse linear models, and more. The Association for

Computing Machinery Conference on Fairness, Accountability, and Transparency (ACM FAccT) was established in 2018 to study transparency and explainability in the context of socio-technical systems, many of which include artificial intelligence. Some techniques allow visualisations of the inputs to which individual software neurons respond to most strongly. Several groups found that neurons can be aggregated into circuits that perform human-comprehensible functions, some of which reliably arise across different networks trained independently. There are various techniques to extract compressed representations of the features of given inputs, which can then be analysed by standard clustering techniques. Alternatively, networks can be trained to output linguistic explanations of their behaviour, which are then directly human-interpretable. Model behaviour can also be explained with reference to training data—for example, by evaluating which training inputs influenced a given behaviour the most, or by approximating its predictions using the most similar instances from the training data. The use of explainable artificial intelligence (XAI) in pain research, specifically in understanding the role of electrodermal activity for automated pain recognition: hand-crafted features and deep learning models in pain recognition, highlighting the insights that simple hand-crafted features can yield comparative performances to deep learning models and that both traditional feature engineering and deep feature learning approaches rely on simple characteristics of the input time-series data.

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Understanding versus trust The goal of explainability to end users of AI systems is to increase trust in the systems, even "address concerns about lack of 'fairness' and discriminatory effects". However, even with a good understanding of an AI system, end users may not necessarily trust the system. In one study, participants were presented with combinations of white-box and black-box explanations, and static and interactive explanations of AI systems. While these explanations served to increase both their self-reported and objective understanding, it had no impact on their level of trust, which remained skeptical. This outcome was especially true for decisions that impacted the end user in a significant way, such as graduate school admissions. Participants judged algorithms to be too inflexible and unforgiving in comparison to human decision-makers; instead of rigidly adhering to a set of rules, humans are able to consider exceptional cases as well as appeals to their initial decision. For such decisions, explainability will not necessarily cause end users to accept the use of decision-making algorithms. We will need to either turn to another method to increase trust and acceptance of decision-making algorithms, or question the need to rely solely on AI for such impactful decisions in the first place. However, some emphasize that the purpose of explainability of artificial intelligence is not

to merely increase users' trust in the system's decisions, but to calibrate the users' level of trust to the correct level. According to this principle, too much or too little user trust in the AI system will harm the overall performance of the human-system unit. When the trust is excessive, the users are not critical of possible mistakes of the system and when the users do not have enough trust in the system, they will not exhaust the benefits inherent in it.

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Participatory budgeting Yang, Hausladen, Peters, Pournaras, Fricker and Helbing present an empirical study of explainability in participatory budgeting. They compared the greedy and the equal shares rules, and three types of explanations: mechanism explanation (a general explanation of how the aggregation rule works given the voting input), individual explanation (explaining how many voters had at least one approved project, at least 10000 CHF in approved projects), and group explanation (explaining how the budget is distributed among the districts and topics). They compared the perceived trustworthiness and fairness of greedy and equal shares, before and after the explanations. They found out that, for MES, mechanism explanation yields the highest increase in perceived fairness and trustworthiness; the second-highest was Group explanation. For Greedy, Mechanism explanation increases perceived trustworthiness but not fairness, whereas Individual explanation increases both perceived fairness and trustworthiness. Group explanation decreases the perceived fairness and trustworthiness.